

Functional (Style)

Quentin Crain

```
nums = [4, 2, 7, 22]
```

New list where each number in `nums` except 7, which is thrown out, is incremented by 3.

Before

```
nums = [4, 2, 7, 22]
```

After

```
def add(a, b):  
    return a + b  
  
nums = [4, 2, 7, 22]  
  
new_nums = []  
  
for num in nums:  
    if num != 7:  
        new_nums.append(add(3, num))  
  
return new_nums
```

3, is invariant and is added cognitive load in `add(3, num)`,
lets get rid of it!

Before

```
def add(a, b):  
    return a + b  
  
nums = [4, 2, 7, 22]  
  
new_nums = []  
  
for num in nums:  
    if num != 7:  
        new_nums.append(add(3, num))  
  
return new_nums
```

After

```
import functools  
  
def add(a, b):  
    return a + b  
add_3 = functools.partial(add, 3)  
  
nums = [4, 2, 7, 22]  
  
new_nums = []  
  
for num in nums:  
    if num != 7:  
        new_nums.append(add_3(num))  
  
return new_nums
```

The condition `num != 7` is invariant so lets make it a function.

Before

```
import functools

def add(a, b):
    return a + b
add_3 = functools.partial(add, 3)

nums = [4, 2, 7, 22]

new_nums = []

for num in nums:
    if num != 7:
        new_nums.append(add_3(num))

return new_nums
```

After

```
import functools
import operator

def add(a, b):
    return a + b
add_3 = functools.partial(add, 3)

ne_7 = functools.partial(operator.ne, 7)

nums = [4, 2, 7, 22]

new_nums = []

for num in nums:
    if ne_7(num):
        new_nums.append(add_3(num))

return new_nums
```

Simple `for` loops should be list comprehensions.

Before

```
import functools
import operator

def add(a, b):
    return a + b
add_3 = functools.partial(add, 3)

ne_7 = functools.partial(operator.ne, 7)

nums = [4, 2, 7, 22]

new_nums = []

for num in nums:
    if ne_7(num):
        new_nums.append(add_3(num))

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After

```
import functools
import operator

def add(a, b):
    return a + b
add_3 = functools.partial(add, 3)

ne_7 = functools.partial(operator.ne, 7)

nums = [4, 2, 7, 22]

new_nums = [add_3(num) for num in nums if ne_7(num)]

return new_nums
```

(This is probably where we stop with Python, but ..

Even explicit looping over `nums` and specifying `num` are invariants, so lets use `filter` then `map`.

Before

```
import functools
import operator

def add(a, b):
    return a + b
add_3 = functools.partial(add, 3)

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nums = [4, 2, 7, 22]

new_nums = [add_3(num) for num in nums if ne_7(num)]

return new_nums
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After

```
import functools
import operator

def add(a, b):
    return a + b
add_3 = functools.partial(add, 3)

ne_7 = functools.partial(operator.ne, 7)

nums = [4, 2, 7, 22]

lnums_without_7 = filter(ne_7, nums)
lnew_nums = map(add_3, lnums_without_7)

return new_nums
```

Curry & Compose!

Before

```
import functools
import operator

def add(a, b):
    return a + b
add_3 = functools.partial(add, 3)

ne_7 = functools.partial(operator.ne, 7)

nums = [4, 2, 7, 22]

nums_without_7 = filter(ne_7, nums)

new_nums = map(add_3, nums_without_7)

return new_nums
```

After

```
import functools
import operator

def add(a, b):
    return a + b
add_3 = functools.partial(add, 3)

ne_7 = functools.partial(operator.ne, 7)

!filter_ne_7 = functools.partial(filter, ne_7)
!map_add_3 = functools.partial(map, add_3)
!filter_ne_7_then_map_add_3 = lambda ns: map_add_3(filter_ne_7(ns))

nums = [4, 2, 7, 22]

new_nums = filter_ne_7_then_map_add_3(nums)

return new_nums
```


Explicit looping is gone, curry & compose, "higher-order" concepts of `map`, `filter`.

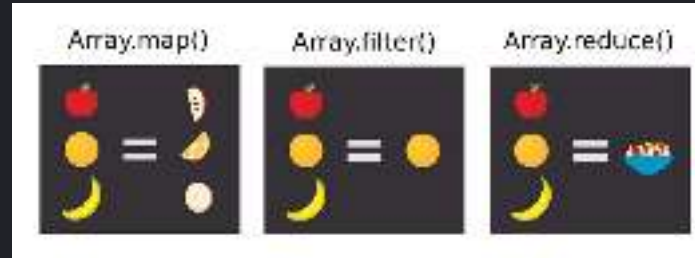
Before

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import functools  
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add_3 = functools.partial(add, 3)  
  
ne_7 = functools.partial(operator.ne, 7)  
  
filter_ne_7 = functools.partial(filter, ne_7)  
map_add_3 = functools.partial(map, add_3)  
filter_ne_7_then_map_add_3 = lambda ns: map_add_3(filter_ne_7(ns))  
  
nums = [4, 2, 7, 22]  
  
new_nums = filter_ne_7_then_map_add_3(nums)  
  
return new_nums
```

`map` , `filter` , `reduce`



Curry (`partial`)

<https://en.wikipedia.org/wiki/Currying>

(Function) Composition

[https://en.wikipedia.org/wiki/Function_composition_\(computer_science\)](https://en.wikipedia.org/wiki/Function_composition_(computer_science))

COMMENTS

QUESTIONS

CONCERNS

Functional is better!

Elixir

```
iex> nums = [4, 2, 7, 22]
[4, 2, 7, 22]

iex> nums
|> Enum.filter(fn n -> n != 7 end)
|> Enum.map(fn n -> n + 3 end)
[7, 5, 25]

# or, with the capture shorthand:
iex> nums
|> Enum.filter(& &1 != 7)
|> Enum.map(& &1 + 3)
[7, 5, 25]
```

Racket

```
> (let [(nums '(4 2 7 22))
        (add-3 (curry + 3))
        (ne-7 (compose not (curry = 7)))]
      (map add-3 (filter ne-7 nums)))
'(7 5 25)

; absurd, or is it??!!?
> (let* [(nums '(4 2 7 22))
         (add-3 (curry + 3))
         (map-add-3 (curry map add-3))
         (ne-7 (compose not (curry = 7)))]
        (filter-ne-7 (curry filter ne-7))
        (filter-ne-7-then-map-add-3 (compose map-add-3 filter-ne-7)))
        (filter-ne-7-then-map-add-3 nums))
'(7 5 25)
```